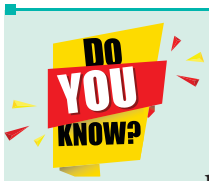
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Table 1.2 Calculation of number of entities in one mole of substance.

S. No.	Name of substance	Mass of the substance taken (gram)	Mass of single atom or molecule (gram)	No. of atoms or molecules = Mass of substance ÷ Mass of single atom or molecule
	(1)	(2)	(3)	(2)÷(3)
1.	Elemental Carbon (C-12)	12	1.9926×10^{-23}	$\frac{12}{1.9926 \times 10^{-23}} = 6.022 \times 10^{23}$
2.	Glucose ($C_6H_{12}O_6$)	180	29.89×10^{-23}	$\frac{180}{29.89 \times 10^{-23}} = 6.022 \times 10^{23}$
3.	Potassim dichromate ($K_2Cr_2O_7$)	294.18	48.851×10^{-23}	$\frac{294.18}{48.851 \times 10^{-23}} = 6.022 \times 10^{23}$
4.	Potassim permang nate ($KMnO_4$)	158.03	26.242×10^{-23}	$\frac{158.03}{26.242 \times 10^{-23}} = 6.022 \times 10^{23}$

From the calculations we come to know that 12 g of carbon-12 contains 6.022×10^{23} carbon atoms and same numbers of molecules are present in 158.03 g of potassium permanganate and 294.18 g of potassium dichromate. Similar to the way we use the term 'dozen' to represent 12 entities, we can use the term 'mole' to represent 6.022×10^{23} entities (atoms or molecules or ions)

One mole is the amount of substance of a system, which contains as many elementary particles as there are atoms in 12 g of carbon-12 isotope. The elementary particles can be molecules, atoms, ions, electrons or any other specified particles.



Gastric acid and antacids:

Antacids are commonly used medicines for treating heartburn and acidity. Do you know the chemistry behind it?

Gastric acid is a digestive fluid formed in the stomach and it contains hydrochloric acid. The typical concentration of the acid in gastric acid is 0.082 M. When the concentration exceeds 0.1 M it causes the heartburn and acidity.

Antacids used to treat acidity contain mostly magnesium hydroxide or aluminium hydroxide that neutralises the excess acid. The chemical reactions are as follows.

$$3\ HCl + Al(OH)_3 \rightarrow AlCl_3 + 3\ H_2O$$

$$2\ HCl + Mg(OH)_2 \rightarrow MgCl_2 + 2\ H_2O$$

From the above reactions we know that 1 mole of aluminium hydroxide neutralises 3 moles of HCl while 1 mole of magnesium hydroxide neutralises 2 moles of HCl.